SUMS OF TWO SQUARES WITH FEW PRIME FACTORS, II: THE CUSPIDAL ERROR IN MEAN SQUARE AND ALMOST ALL N

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ABSTRACT. Let m, m' be coprime odd integers, $\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z)$, $G = G_{m,m'}$ its cuspidal part and $L = 4mm'$, so that $a_G(N) = T(m, m'; N) - M(m, m'; N)$ is the error term in the number of representations of N by $m(x^2 + y^2) + m'(z^2 + w^2)$. We prove the mean-square bound $\sum_{N\le Y} |a_G(N)|^2 \le 4\pi e^2\, Y(1 + 54Y/L)\|G\|^2$, which with Waibel's norm bound gives $\sum_{N\le Y} |T - M|^2 \ll Y^2(mm')^{-1/2+\varepsilon}$ for $Y \ge mm'$: on average over N the error is $N^{1/2}(mm')^{-1/4}$, the size observed numerically in Part I. It follows that the weighted sequence $r(n)r(N-n)$ has level of distribution $2/3$ for all but $O(X^{1-\delta})$ integers $N \equiv 2 \pmod 4$ up to X, and for these N the sieve of Part I gives $\gg N/(\log N)^2$ integers $n < N$ with n, $N-n$ sums of two squares, $\Omega(n(N-n)) \le 6$ and $\max(\Omega(n), \Omega(N-n)) \le 4$. We reformulate the uniform conjecture of Part I as a near-orthogonality statement between $\Theta_{m,m'}$ and the Poincaré series, show that it holds on average in a strong form, and record numerical evidence that $\|G_{m,m'}\|^2 \approx 3\,(mm')^{1/4}$, smaller than the general bound.

1. RESULTS

We keep the notation of [1], referred to as Part I. Thus $r(n)$ is the number of representations of n as a sum of two squares, m, m' are coprime odd positive integers,

$$T(m, m'; N) = \sum_{ma+m'b=N} r(a)r(b), \qquad L = 4mm',$$

$$\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z) = \sum_{N\ge 0} T(m, m'; N)e(Nz) \in M_2(\Gamma_0(L)),$$

$\Theta_{m,m'} = E + G$ is the decomposition into the Eisenstein part, whose coefficients are the Siegel main terms $M(m, m'; N)$, and the cuspidal part $G = G_{m,m'} \in S_2(\Gamma_0(L))$, so that

$$a_G(N) = T(m, m'; N) - M(m, m'; N) \qquad \text{for every } N \ge 1.$$

The Petersson norm is $\|G\|^2 = \int_{\Gamma_0(L)\backslash\mathbb{H}} |G|^2 \,\mathrm{d}x\,\mathrm{d}y$, and Lemma 2.1 of Part I (Waibel's theorem) gives $\|G\| \ll (mm')^{1/4+\varepsilon}$. Part I proves $a_G(N) \ll d(N)N^{1/2}(mm')^{1/4+\varepsilon}$ for $(N, mm') = 1$, $v_2(N) \le 1$, and conjectures (Conjecture 5.1 there) that the factor $(mm')^{1/4}$ can be removed. The first result of this note is that the conjecture holds in mean square over N, with room to spare.

Theorem 1.1 (Mean square)**.** *Let $L \ge 1$, $G \in S_2(\Gamma_0(L))$ and $Y \ge 1$. Then*

$$\sum_{n\le Y} |a_G(n)|^2 \;\le\; 4\pi e^2\, Y\Big(1 + \frac{54\,Y}{L}\Big)\|G\|^2. \tag{1}$$

In particular, for coprime odd m, m', $G = G_{m,m'}$ and $Y \ge mm'$,

$$\sum_{N\le Y} \big|T(m, m'; N) - M(m, m'; N)\big|^2 \;\ll_\varepsilon\; Y^2\,(mm')^{-1/2+\varepsilon}, \tag{2}$$

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and hence, since $\sum_{Y/2<N\le Y}|a_G(N)|^2/N \le (2/Y)\sum_{N\le Y}|a_G(N)|^2$, *the root mean square of* $(T-M)/\sqrt{N}$ *over* $Y/2<N\le Y$ *is* $\ll (mm')^{-1/4+\varepsilon}$.

The bound (2) is what the data of Part I, Table 3, show (the root mean square there decreases with mm'; see Section 5 for a closer look). The proof of Theorem 1.1 is a classical covering argument and does not use the Petersson formula; the dependence on the level is exactly the factor Y/L, i.e. the volume of $\Gamma_0(L)\backslash\mathbb{H}$.

Averaging over N is natural for the arithmetic application, since the sieve of Part I is applied to one N at a time and an exceptional set of N can be tolerated. Let, as in Part I, $N\equiv 2 \pmod 4$, $\mathcal{A}=\{n\le N:\ (n,N)=1\}$, $w(n)=r(n)r(N-n)$, and for coprime square-free d,e with $(de,2N)=1$ let $R(d,e)=R(d,e;N)$ be the remainder in

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n,\ e\mid N-n}} w(n) = X_N\, h(d)h(e) + R(d,e;N)$$

(this is $S(d,e)=M(d,e)+R(d,e)$ of Part I, Section 3; we write X_N for the main term X there, to avoid a clash with the length X of the range of N below).

Theorem 1.2 (Level of distribution 2/3 for almost all N)**.** *Let* $0<\varepsilon<1/20$, $X\ge X_0(\varepsilon)$ *and* $D=X^{2/3-2\varepsilon}$. *Then*

$$\sum_{\substack{X/2<N\le X\\ N\equiv 2\ (4)}}\ \sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \ll_\varepsilon\ X^{2-\varepsilon/2}.$$

Consequently, for all but $O_\varepsilon(X^{1-\varepsilon/4})$ *integers* $N\equiv 2 \pmod 4$ *in* $(X/2,X]$,

$$\sum_{\substack{de\le N^{2/3-2\varepsilon}\\ (d,e)=1,\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \le\ N^{1-\varepsilon/5}. \tag{3}$$

For comparison, Theorem 3.2 of Part I gives (3) with 2/5 in place of 2/3 for every N, and Conjecture 5.1 of Part I would give 1/2 for every N. The sieve of Part I, Section 4, uses the level of distribution only through (3); running it at $\theta=2/3$ gives the following.

Corollary 1.3. *For all but* $O(X^{1-\delta})$ *integers* $N\equiv 2 \pmod 4$ *in* $(X/2,X]$, *with an absolute* $\delta>0$,

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 6,\ \max(\Omega(n),\Omega(N-n))\le 4\big\}$$
$$\ge\ 0.13\,\mathfrak{S}_r(N)\,\frac{N}{(\log N)^2},\qquad \mathfrak{S}_r(N)=\prod_{\substack{p\mid N\\ p>2}}\frac{p-1}{p-2},$$

the integers counted being coprime to N *and products of distinct primes* $\equiv 1 \pmod 4$ *exceeding* $N^{3/50}$.

The pair $(6,4)$ should be compared with $(9,6)$ for every N (Part I, Theorem 1.1) and $(8,5)$ for every N under Conjecture 5.1 of Part I. The certified sieve parameters are $\alpha=3/50$, $\lambda=1/3$, $\beta_1=51/100$ (Section 3).

Section 4 returns to the uniform conjecture. Writing P_N for the N-th Poincaré series of $S_2(\Gamma_0(L))$, one has $a_G(N)=4\pi N\langle G,P_N\rangle$, so that

$$|a_G(N)|^2 = 4\pi N\,\Delta_L(N,N)\,\|G\|^2\cos^2\angle(G,P_N),$$

where $\Delta_L(N,N)$ is the Petersson density of Part I, (11). Theorem 1.1 says that the average over $N\le Y$ of $\Delta_L(N,N)\cos^2\angle(G,P_N)$ is $O(1/L)$ for $Y\ge L$, the value one would get from

a random vector in the $\asymp L$-dimensional space $S_2(\Gamma_0(L))$; Conjecture 5.1 of Part I asks only that no single N exceed this average by more than a factor $N^{\varepsilon}L/\|G\|^2$, which is $\gg (mm')^{1/2-\varepsilon}$ by Waibel's bound and about $(mm')^{3/4}$ if the numerical estimate below is right. Section 5 contains the numerical evidence for Theorem 1.1 and the estimate $\|G_{m,m'}\|^2 \approx 3(mm')^{1/4}$ from the Rankin–Selberg asymptotic.

2. Proof of Theorem 1.1

Let $G(z) = \sum_{n\ge1} a(n)e(nz) \in S_2(\Gamma_0(L))$ and put $F(z) = y^2|G(z)|^2$, $z = x+iy$. Since $G(\gamma z) = (cz+d)^2 G(z)$ and $\operatorname{Im}(\gamma z) = y/|cz+d|^2$ for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma_0(L)$, the function F is $\Gamma_0(L)$-invariant, and with the invariant measure $\mathrm{d}\mu = y^{-2}\mathrm{d}x\,\mathrm{d}y$ we have $\|G\|^2 = \int_{\mathcal F} F\,\mathrm{d}\mu$ for any fundamental domain $\mathcal F$ of $\Gamma_0(L)$.

Let $y_0 = 1/(4\pi Y)$ and $R = \{x+iy:\ 0\le x<1,\ y_0 \le y \le 2y_0\}$ (half-open in x, so that each orbit of the translations $z\mapsto z+b$ meets R at most once). By Parseval, $\int_0^1 |G(x+iy)|^2\,\mathrm{d}x = \sum_n |a(n)|^2 e^{-4\pi n y}$, hence

$$\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \sum_n |a(n)|^2 \int_{y_0}^{2y_0} e^{-4\pi n y}\,\mathrm{d}y \ \ge\ \frac{e^{-2}}{4\pi Y}\sum_{n\le Y}|a(n)|^2,$$

because $4\pi n y \le 8\pi Y y_0 = 2$ for $n\le Y$, $y\le 2y_0$. On the other hand $\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \int_R F\,\mathrm{d}\mu$, and

$$\int_R F\,\mathrm{d}\mu = \sum_{\gamma\in\Gamma_0(L)/\{\pm1\}} \int_{R\cap\gamma\mathcal F} F\,\mathrm{d}\mu = \int_{\mathcal F} F(w)\,\#\{\gamma:\ \gamma w\in R\}\,\mathrm{d}\mu(w) \ \le\ \mu_R\,\|G\|^2,$$

where $\mu_R = \sup_{z\in R}\#\{\gamma\in\Gamma_0(L)/\{\pm1\}:\ \gamma z\in R\}$ (if $\#\{\gamma:\gamma w\in R\}\neq 0$ for some $w\in\mathcal F$, pick γ_0 with $z=\gamma_0 w\in R$; then $\gamma\mapsto\gamma\gamma_0^{-1}$ is a bijection from $\{\gamma:\gamma w\in R\}$ onto $\{\gamma:\gamma z\in R\}$, so the count at w is at most μ_R). It remains to bound μ_R. Fix $z=x+iy\in R$ and let $\gamma = \left(\begin{smallmatrix} a & b\\ c & d\end{smallmatrix}\right)\in\Gamma_0(L)$ with $\gamma z\in R$.

- If $c=0$, then $\gamma = \pm\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right)$ and $\gamma z = z+b$, so $x+b\in[0,1)$ forces $b=0$: one element.
- If $c\neq 0$, we may take $c>0$, since γ and $-\gamma$ are the same element of $\Gamma_0(L)/\{\pm1\}$. Then $L\mid c$, and $\operatorname{Im}(\gamma z) = y/|cz+d|^2 \in [y_0,2y_0]$ together with $y\in[y_0,2y_0]$ gives $|cz+d|^2\in[1/2,2]$. Since $|cz+d|^2 = (cx+d)^2+c^2y^2$, this forces $c^2y^2\le 2$, hence $c\le\sqrt{2}/y_0 = 4\sqrt{2}\,\pi Y$, and $|cx+d|\le\sqrt{2}$. The number of positive multiples c of L with $c\le 4\sqrt{2}\pi Y$ is $\lfloor 4\sqrt{2}\pi Y/L\rfloor \le 4\sqrt{2}\pi\,Y/L$; for each c there are at most 3 integers d with $|cx+d|\le\sqrt{2}$; and for each pair (c,d) the elements of $\Gamma_0(L)$ with bottom row (c,d) are $\left(\begin{smallmatrix}1 & k\\ 0 & 1\end{smallmatrix}\right)\gamma_0$, $k\in\mathbb Z$, whose images $\gamma_0 z + k$ have real part in $[0,1)$ for exactly one k. This gives at most $4\sqrt{2}\pi\cdot 3\,Y/L = 12\sqrt{2}\pi\,Y/L < 54\,Y/L$ elements.

Hence $\mu_R \le 1+54Y/L$, and

$$\sum_{n\le Y}|a(n)|^2 \le 4\pi e^2 Y\int_R F\,\mathrm{d}\mu \le 4\pi e^2\,Y\Big(1+\frac{54Y}{L}\Big)\|G\|^2,$$

which is (1). For $G=G_{m,m'}$, $L=4mm'$ and $Y\ge mm'$ the bracket is $\le 1+54Y/(4mm') \le 15Y/(mm')$, and Lemma 2.1 of Part I (which is stated for all coprime odd m,m', square factors included, the level of $Q_{m,m'}$ being $4mm'$ in every case) gives (2). □

Remark 2.1. The argument is the classical one for bounding $\sum_{n\le Y}|a(n)|^2$ by the norm (compare [2, Chapter 5]); its only input is the invariance of $y^2|G|^2$. The strip $y_0\le y\le 2y_0$ with $y_0 = 1/(4\pi Y)$ is chosen so that the exponential factor in Parseval's identity stays at e^{-2} for all $n\le Y$; the strip $1/Y\le y\le 2/Y$ would give the same argument with $e^{8\pi}$ in place of $4\pi e^2$. For $Y\ge L$ the factor Y^2/L in (1) is of the correct order of magnitude: by the Rankin–Selberg method, $\sum_{n\le Y}|a(n)|^2/n \sim 16\pi^2\|G\|^2 Y/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb H)$ with $\operatorname{vol} = \frac{\pi}{3}L\prod_{p\mid L}(1+1/p)$ (measure

$\mathrm{d}\mu$), so that $\sum_{n\le Y}|a(n)|^2 \sim 8\pi^2\|G\|^2 Y^2/\operatorname{vol}$ for fixed G as $Y\to\infty$. Theorem 1.1 makes this uniform in the level, at the cost of the constant.

3. Proof of Theorem 1.2 and Corollary 1.3

3.1. **The remainder on average.** We follow the proof of Theorem 3.2 of Part I, replacing the uniform bound for $|T-M|$ by Theorem 1.1 after an application of Cauchy–Schwarz over N. Recall from Part I, (15), that for $N\equiv 2 \pmod 4$ and coprime square-free d,e with $(de,2N)=1$,

$$R(d,e;N)=\sum_{t\mid N^\flat}\mu(t)\sum_{m\mid d^2}\sum_{m'\mid e^2}c_d(m)c_e(m')\sum_{\substack{t_1\mid t^2\\ t\mid t_1}}c_t(t_1)\big(a_G(N)-a_G(N/2)\big),\qquad G=G_{t_1m,m'}, \tag{4}$$

where $N^\flat$ is the product of the odd primes dividing N, $m=ds$, $m'=es'$ with $s\mid d$, $s'\mid e$, and likewise $t_1=tu$ with $u\mid t$ (the coefficients $c_d(m)$, $c_t(t_1)$ of Part I, (14), vanish unless $m=ds$ with $s\mid d$, resp. $t_1=tu$ with $u\mid t$; in particular $t\mid t_1$); the parameters t_1m,m' of the form $Q_{t_1m,m'}$ are coprime, odd and cube-free (not necessarily square-free; Lemma 2.1 of Part I covers this case), $\tilde m\tilde m' := t_1mm' = t_1ss'\cdot de$, and the coefficients c satisfy $|c_d(m)c_e(m')c_t(t_1)|\le 3^{\omega(d)+\omega(e)+\omega(t)}$. (For the reader's convenience: c_d is the multiplicative coefficient of Part I, (14), with $c_p(p)=2$, $c_p(p^2)=-1$ for a split prime p and $c_q(q)=0$, $c_q(q^2)=1$ for an inert prime q; it encodes the identities $r(n)=2r(n/p)-r(n/p^2)$ and $r(n)=r(n/q^2)$, which turn the condition $d\mid n$ into a combination of theta series $\theta^2(mz)$. The sum over t is the Möbius inversion of the condition $(n,N)=1$, and $a_G(N)-a_G(N/2)$ selects the odd n.) Let $\eta=\varepsilon/10$. For $N\le X$ with $X\ge X_0(\varepsilon)$ we have, as in (F1)–(F3) of Part I: $d_3(de)\,3^{\omega(d)+\omega(e)+\omega(t)}\,4^{\omega(N)}d(N)^2\le X^\eta$; for fixed (d,e,N) the number of tuples (s,s',t,t_1) is at most X^η; and $\sum_{k\le D}d(k)/k\le X^\eta$, $\sum_{k\le D}d(k)k^{-1/4}\le D^{3/4}X^\eta$. We also use the two bounds for a single term,

$$|a_G(N')|\le T(\tilde m,\tilde m';N')+M(\tilde m,\tilde m';N')\ll X^{2\eta}\Big(\frac{X}{\tilde m\tilde m'}+1\Big)\qquad (N'\le X), \tag{5}$$

$$\sum_{n\le X}|a_G(n)|\le X^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}\ll X\Big(1+\frac{X}{L}\Big)^{1/2}L^{1/4+\eta}, \tag{6}$$

where $G=G_{\tilde m,\tilde m'}$ and $L=4\tilde m\tilde m'$, the first being (20) of Part I, the second Cauchy–Schwarz, (1) and Lemma 2.1 of Part I; for $L\le X$ the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/4+\eta}$. Let Σ denote the double sum in Theorem 1.2. Inserting (4) and splitting the tuples according to the size of t_1ss':

Tuples with $t_1ss'>X^\varepsilon$. For each N and each tuple, (5) with $\tilde m\tilde m'>X^\varepsilon de$ gives $|a_G(N')|\ll X^{2\eta}(X^{1-\varepsilon}/(de)+1)$. Summing over the $\le X^\eta$ tuples, over $N\le X$ and over $de\le D$ with the weights $d_3(de)\cdot 3^\omega\le X^\eta$, these tuples contribute

$$\ll X^{1+4\eta}\sum_{k\le D}d(k)\Big(\frac{X^{1-\varepsilon}}{k}+1\Big)\ll X^{2-\varepsilon+5\eta}+X^{1+5\eta}D\ll X^{2-\varepsilon/2}.$$

Tuples with $t_1ss'\le X^\varepsilon$. Now $t\le t_1\le X^\varepsilon$ (as $t\mid t_1$), so t runs over at most X^ε values, t_1 over at most $d(t^2)\le X^\eta$ values, and (s,s') over at most $d(d)d(e)\le X^\eta$ values; the forms $G=G_{t_1m,m'}$ have level $L=4t_1ss'de\le 4X^\varepsilon D\le X$, so that the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/4+\eta}$. For a fixed tuple and fixed (d,e), we drop the condition $t\mid N$ except for the count: the number

of $N \le X$ with $t \mid N$ is $\le X/t$, and Cauchy–Schwarz gives

$$\sum_{\substack{N\le X\\ t\mid N}} \big(|a_G(N)| + |a_G(N/2)|\big) \le 2\Big(\frac{X}{t}\Big)^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}$$
$$\ll \Big(\frac{X}{t}\Big)^{1/2} X\,(\tilde m\tilde m')^{-1/4+\eta} \le X^{3/2+\eta}\,t^{-1/2}(de)^{-1/4},$$

using (1) in the form $\sum_{n\le X}|a_G(n)|^2 \ll X^2\|G\|^2/L$ and $(\tilde m\tilde m')^{-1/4+\eta} \le X^{\eta}(de)^{-1/4}$. Summing over $t \le X^{\varepsilon}$ ($\sum_t t^{-1/2} \le 2X^{\varepsilon/2}$), over the other tuple variables ($\le X^{2\eta}$), and over $de \le D$ with the weights $\le X^{\eta}$,

$$\ll X^{3/2+4\eta+\varepsilon/2}\sum_{k\le D} d(k)k^{-1/4} \ll X^{3/2+5\eta+\varepsilon/2}D^{3/4} = X^{2-\varepsilon+5\eta} = X^{2-\varepsilon/2},$$

since $D^{3/4} = X^{1/2-3\varepsilon/2}$. Both contributions are $\ll X^{2-\varepsilon/2}$, which proves the first assertion. (The margin is small but positive: the choice $D = X^{2/3-2\varepsilon}$ saves $X^{-3\varepsilon/2}$ in $D^{3/4}$, of which the sum over t costs $X^{\varepsilon/2}$ and the divisor-type factors cost $X^{5\eta} = X^{\varepsilon/2}$; this is why $\eta = \varepsilon/10$.)

For the second, let $S(N)$ be the inner sum with $D = X^{2/3-2\varepsilon}$. The number of $N \in (X/2, X]$, $N \equiv 2 \pmod 4$, with $S(N) > X^{1-\varepsilon/4}$ is at most $\Sigma/X^{1-\varepsilon/4} \ll X^{1-\varepsilon/4}$. For the remaining N, since $N^{2/3-2\varepsilon} \le D$ and $X^{1-\varepsilon/4} \le (2N)^{1-\varepsilon/4} \le N^{1-\varepsilon/5}$ for large X, (3) holds. □

Remark 3.1. The exponent $2/3$ is the balance $X^{3/2}D^{3/4} = X^2$ between the mean-square bound $X^{3/2}(de)^{-1/4}$ for the sum over N of one remainder and the number D of moduli, exactly as $2/5$ in Part I is the balance $N^{1/2}D^{1/4}\cdot D = N$ for a single N. Any improvement of the exponent $1/4$ in Waibel's bound, say $\|G_{m,m'}\| \ll (mm')^{\beta+\varepsilon}$, gives the level $1/(2(1+\beta))$ for every N (Part I, Remark 3.3) and, through the balance $X^{3/2}D^{1/2+\beta} = X^2$, the level $1/(1+2\beta)$ for almost all N; for $\beta = 1/4$ these are $2/5$ and $2/3$, and for $\beta = 1/8$ (Section 5) they would be $4/9$ and $4/5$.

Remark 3.2. The exceptional set $X^{1-\varepsilon/4}$ comes from Markov's inequality applied to a first moment of $S(N)$, which in turn rests on the second moment of $a_G(N)$. A bound for a higher moment, $\sum_{N\le Y}|a_G(N)|^{2k} \ll Y^{k+1}(mm')^{-k/2+\varepsilon}$, would shrink the exceptional set to $X^{1-k\varepsilon'}$; we have no such bound for $k \ge 2$, and do not pursue this.

3.2. **The sieve at level $2/3$.** Proposition 4.3 of Part I and its proof use the level of distribution only through the bound (3) (there with $2/5$); they hold verbatim, for a given N, with $\theta = 2/5$ replaced by any $\theta \in (0,1)$ for which (3) holds with $N^{\theta-\varepsilon}$ in place of $N^{2/3-2\varepsilon}$, provided $\alpha < \beta_1 < \theta$. Lemmas 4.1 and 4.2 of Part I do not involve the level at all. Hence, for every N satisfying (3), the conclusion of Proposition 4.3 holds with $\theta = 2/3$ (the ε there being 2ε here), and the proof of Theorem 1.1 of Part I goes through with the parameters certified for $\theta = 2/3$. The script `certify_modular.py` of Part I, extended by the argument `2/3` in the repository of this paper, certifies in interval arithmetic

$$\theta = \tfrac{2}{3}, \qquad \alpha = \tfrac{3}{50}, \qquad \lambda = \tfrac{1}{3}, \qquad \beta_1 = \tfrac{51}{100},$$
$$\tfrac{1}{\lambda} + \tfrac{2}{\beta_1} = \tfrac{353}{51} < 7, \qquad \tfrac{1}{\lambda} + \tfrac{1}{\beta_1} = \tfrac{253}{51} < 5, \qquad L - \lambda R \ge 0.036741$$

(with $L \ge 0.998974$, $R \le 2.886700$, $\theta_A = 0.3330$; a second set $\alpha = 29/500$, $\beta_1 = 13/25$ gives $L - \lambda R \ge 0.012502$; the output files are listed at the end of the paper). By Lemma 4.2 of Part I this gives $\Omega(n(N-n)) \le 6$ and $\max(\Omega(n), \Omega(N-n)) \le 4$, and the weight of each n counted is $w(n) = 16\cdot 2^{\Omega(n(N-n))} \le 2^{10}$. As in the proof of Theorem 1.1 of Part I, the number of integers counted is at least

$$\big(0.036741 + o(1)\big)\,2^{-10}\cdot 8\,\mathfrak{S}_r(N)\,c_{2h}\,\frac{N}{\alpha^2(\log N)^2} \;\ge\; 0.13\,\mathfrak{S}_r(N)\frac{N}{(\log N)^2},$$

since $0.036741 \cdot 2^{-10} \cdot 8 \cdot 1.66485 \cdot (50/3)^2 = 0.1327\ldots$. The exceptional set is that of Theorem 1.2 with, say, $\varepsilon = 1/40$, so $\delta = 1/160$ is admissible. □

Remark 3.3. At the level 1/2 the same sieve gives $(8, 5)$ (Part I, Remark 4.5), at 2/3 it gives $(6, 4)$. The floating-point scan of `modular_sieve_scan.py` at $\theta = 2/3$ shows that $(6, 4)$ is the limit of the method at this level: the best value of $1/\lambda + 2/\beta_1$ is about 6.8 and that of $1/\lambda + 1/\beta_1$ about 4.65, so neither $(5, \cdot)$ nor $(\cdot, 3)$ is reachable with these weights. Corollary 1.3 is an "almost all N" statement with a power-saving exceptional set; it says nothing about any individual N, and in particular nothing about the integers N of Theorem 1.1 of Part I beyond what is proved there.

4. The uniform conjecture as a statement about Poincaré series

Let $L = 4mm'$ and let $P_N \in S_2(\Gamma_0(L))$, $N \geq 1$, be the Riesz representative of the N-th coefficient: the unique cusp form with

$$\langle f, P_N \rangle = \frac{a_f(N)}{4\pi N} \qquad \text{for all } f \in S_2(\Gamma_0(L)), \tag{7}$$

i.e. $P_N = (4\pi N)^{-1} \sum_{f \in \mathcal{B}} \overline{a_f(N)}\, f$ for an orthonormal basis $\mathcal{B}$. (This is the N-th Poincaré series; for weight 2 the series itself converges only conditionally, which is why we define P_N by (7).) By (11) of Part I,

$$\|P_N\|^2 = \frac{a_{P_N}(N)}{4\pi N} = \frac{1}{(4\pi N)^2} \sum_{f \in \mathcal{B}} |a_f(N)|^2 = \frac{\Delta_L(N, N)}{4\pi N}.$$

Hence, for $G = G_{m,m'}$,

$$|a_G(N)|^2 = (4\pi N)^2 |\langle G, P_N \rangle|^2 = 4\pi N\, \Delta_L(N, N)\, \|G\|^2 \cos^2 \angle(G, P_N), \tag{8}$$

where $\cos \angle(G, P_N) = |\langle G, P_N \rangle| / (\|G\|\, \|P_N\|)$. This identity separates the three quantities that enter any bound for $a_G(N)$.

(i) $\|G\|^2$: controlled by Waibel's theorem, $\ll (mm')^{1/2+\varepsilon}$; numerically $\approx 3(mm')^{1/4}$ (Section 5).
(ii) $\Delta_L(N, N)$: the Petersson density; $\leq C'$ for $N \leq L^{1/2}$ (Part I, Lemma 2.2, Step 4), and $1 + O(N^{1/2+\varepsilon}/L)$ in general, which is all that is known for $N \gg L$.
(iii) $\cos^2 \angle(G, P_N)$: the alignment of the theta series with the Poincaré series. Lemma 2.2 of Part I bounds it by 1 (after moving the coprime part of N into a Hecke operator, which replaces P_N by P_{n_2} with $n_2 \in \{1, 2\}$ and G by $T_{n_1} G$).

Theorem 1.1 is a statement about (iii) on average. Inserting (8) into (1) and restricting to $N \in (Y/2, Y]$,

$$\frac{1}{Y/2} \sum_{Y/2 < N \leq Y} \Delta_L(N, N) \cos^2 \angle(G, P_N) \;\leq\; 4e^2 \Big(\frac{1}{Y} + \frac{54}{L} \Big) \;\ll\; \frac{1}{L} \qquad (Y \geq L). \tag{9}$$

Since $\dim S_2(\Gamma_0(L)) \asymp L$, the right-hand side is the value of $\cos^2$ for a unit vector in general position: on average over N, the theta series is no more aligned with the Poincaré series than a random cusp form of level L would be. In these terms, Conjecture 5.1 of Part I reads

$$\Delta_L(N, N) \cos^2 \angle(G_{m,m'}, P_N) \;\ll_\varepsilon\; \frac{N^\varepsilon}{\|G_{m,m'}\|^2} \qquad \big((N, mm') = 1,\ v_2(N) \leq 1\big), \tag{10}$$

and since $\|G\|^2 \ll (mm')^{1/2+\varepsilon} \ll L^{1/2+\varepsilon}$, the right-hand side of (10) exceeds the average (9) by a factor of at least $L^{1/2-\varepsilon}$. Thus the conjecture does not ask for square-root cancellation at every N, only that no N be aligned with G more than $L^{1/2}$ times the average; what is missing is a bound for the alignment that holds for every N rather than on average.

What such a bound requires. Expanding both sides in an orthogonal basis $\mathcal{B}^*$ of Hecke eigenforms as in Part I, Section 5.5,

$$\langle G, P_N\rangle = \sum_{\varphi\in\mathcal{B}^*} \frac{\langle G,\varphi\rangle\langle\varphi,P_N\rangle}{\|\varphi\|^2} = \frac{1}{4\pi N}\sum_{\varphi\in\mathcal{B}^*} \frac{\langle\Theta,\varphi\rangle\, a_\varphi(N)}{\|\varphi\|^2}$$

(by (7), and $\langle G,\varphi\rangle = \langle\Theta,\varphi\rangle$ since $E\perp S_2$; the coefficients $a_\varphi(N)$ are real for a suitably chosen $\mathcal{B}^*$, but no conjugation occurs in any case), the Cauchy–Schwarz bound $|\langle G,P_N\rangle|\le\|G\|\|P_N\|$ is attained only if the vector $(\langle\Theta,\varphi\rangle/\|\varphi\|)_\varphi$ is proportional to $(a_\varphi(N)/\|\varphi\|)_\varphi$. The missing lemma is therefore a statement about the *correlation* between the two families $\langle\Theta_{m,m'},\varphi\rangle$ (fixed by m,m', independent of N) and $a_\varphi(N)$ (independent of m,m' beyond the level), as φ runs over $\mathcal{B}^*$:

$$\Big|\sum_{\varphi\in\mathcal{B}^*}\frac{\langle\Theta,\varphi\rangle\, a_\varphi(N)}{\|\varphi\|^2}\Big| \ll_\varepsilon N^{1/2+\varepsilon} \qquad \text{uniformly in } m,m' \text{ and } N \text{ with } (N,mm')=1. \tag{11}$$

Theorem 1.1 shows that (11) holds in mean square over N with the stronger right-hand side $N^{1/2}(mm')^{-1/4+\varepsilon}$. We do not know how to prove (11) for individual N. The natural route is to open the coefficients $\langle\Theta,\varphi\rangle$ by an approximate functional equation, of length about the conductor $\asymp mm'$ of the pair of central values in Part I, (5.5), and to average over φ with the Petersson formula; the sum over φ then produces a diagonal term and sums of Kloosterman sums $S(aN,b;c)$ with $c\equiv 0 \pmod L$ and a,b up to about mm'. In the range relevant for the level of distribution, $N\ge(mm')^2\asymp L^2$, the twist by N is longer than the level, and Weil's bound for the individual Kloosterman sums does not suffice; one would need cancellation in the sum over c (through the Kuznetsov formula) or over a. This is the obstacle described in Part I, Section 5.5, and we have nothing to add to it here.

5. Numerical evidence

The script `mean_square_check.py` computes, with the tables and the FFT convolution of `sup_error_scan.py` of Part I, the sums $S_2(Y)=\sum_{N\in\mathcal{N}_Y}|a_G(N)|^2$ over the admissible set

$$\mathcal{N}_Y = \{N\le Y:\ (N,mm')=1,\ v_2(N)\le 3\},$$

on which the main term of that script is exact, for $Y=2^{21}$ and $Y=2^{20}$, and the estimate

$$\|G\|^2_{\mathrm{est}} = \frac{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})}{16\pi^2}\cdot\frac{1}{\#\mathcal{N}_Y}\sum_{N\in\mathcal{N}_Y}\frac{|a_G(N)|^2}{N}$$

suggested by the Rankin–Selberg asymptotic of Remark 2.1. Replacing $1/Y$ by $1/\#\mathcal{N}_Y$ assumes that the mean square of $|a_G(N)|^2/N$ over the excluded N (those with $(N,mm')>1$ or $v_2(N)\ge 4$) is the same as over $\mathcal{N}_Y$; this is immaterial when mm' has only large prime factors, but introduces a bias of unknown sign for $(1,5)$, $(9,1)$, $(3,5)$ and $(1,15015)$, where 25%, 38%, 50% and 64% of the N are excluded. Table 1 shows the result at $Y=2^{21}$, and Figure 1 the dependence on Y and on mm'.

(m, m')	mm'	$\#N$	$S_2(Y)/S_2(Y/2)$	$\|G\|^2_{\mathrm{est}}$	$\|G\|^2_{\mathrm{est}}/(mm')^{1/2}$	$\|G\|^2_{\mathrm{est}}/(mm')^{1/4}$
$(1, 5)$	5	1572864	4.001	0.74	0.33	0.50
$(9, 1)$	9	1310720	4.001	1.47	0.49	0.85
$(3, 5)$	15	1048576	4.001	3.44	0.89	1.75
$(1, 101)$	101	1946614	4.000	7.95	0.79	2.51
$(13, 29)$	377	1752262	4.001	13.09	0.67	2.97
$(1, 1009)$	1009	1964131	3.976	17.96	0.57	3.19
$(61, 67)$	4087	1904986	3.961	24.47	0.38	3.06
$(101, 103)$	10403	1927715	3.948	29.81	0.29	2.95
$(1, 15015)$	15015	754220	4.359	36.62	0.30	3.31
$(1009, 1013)$	1022117	1962192	4.559	46.87	0.046	1.47

TABLE 1. Mean square of the cuspidal error over the admissible $N \le Y = 2^{21}$ (`results/mean_square_check_x21.txt`). The ratio $S_2(Y)/S_2(Y/2) \approx 4$ is the growth Y^2 of Theorem 1.1. For the first three pairs and for $(1, 15015)$ a large proportion of the N (25% to 64%) is excluded by the coprimality condition, and for the last pair $Y < L$, so that neither (1) in its Y^2/L form nor the Rankin–Selberg asymptotic applies; these rows are shown for completeness.

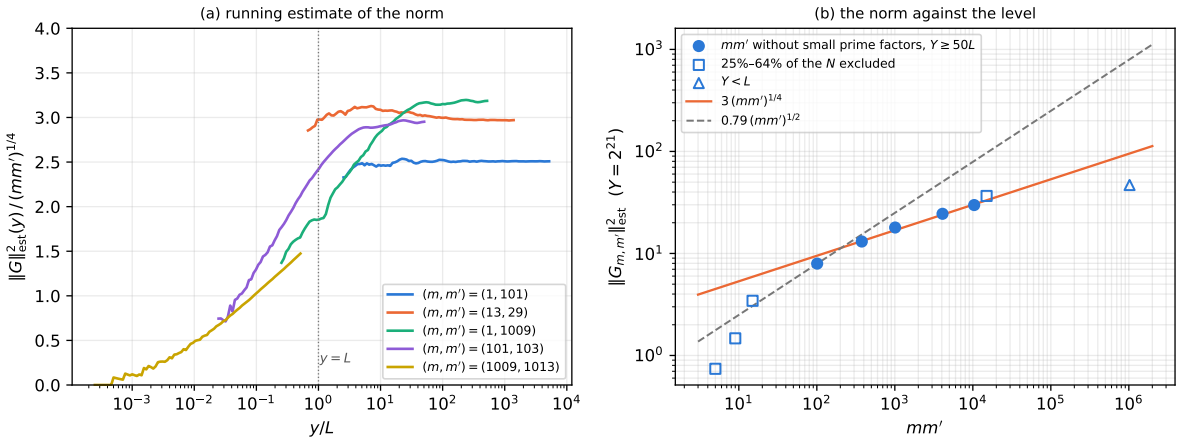


FIGURE 1. (a) The running estimate $\|G\|^2_{\mathrm{est}}(y)$, computed over the admissible $N \le y$, divided by $(mm')^{1/4}$, against y/L on a logarithmic scale, for five pairs (m, m'); the plateau for $y \gg L$ is the growth Y^2 of Theorem 1.1 and the Rankin–Selberg asymptotic of Remark 2.1. (b) $\|G_{m,m'}\|^2_{\mathrm{est}}$ at $Y = 2^{21}$ against mm', both axes logarithmic; filled circles: the five pairs of Observation 5.1; open squares: pairs with 25%–64% of the N excluded; triangle: $(1009, 1013)$, for which $Y < L$. The lines are $3(mm')^{1/4}$ and $0.79(mm')^{1/2}$ (the latter through the point $(1, 101)$, the growth allowed by Waibel's bound). Script `plot_mean_square.py`; floating point.

Observation 5.1. For the five pairs with $10^2 \le mm' \le 1.1 \cdot 10^4$, $Y \ge 50L$ and mm' free of small prime factors, the estimate $\|G_{m,m'}\|^2_{\mathrm{est}}$ lies between $2.5\,(mm')^{1/4}$ and $3.2\,(mm')^{1/4}$ (and $(1, 15015)$, with its bias, gives 3.3), while $\|G\|^2_{\mathrm{est}}/(mm')^{1/2}$ decreases from 0.79 to 0.29 over the same range. The data therefore suggest

$$\|G_{m,m'}\|^2 \asymp (mm')^{1/4}, \qquad \text{i.e.} \qquad \|G_{m,m'}\| \asymp (mm')^{1/8},$$

rather than the bound $(mm')^{1/4+\varepsilon}$ of Lemma 2.1 of Part I. This corrects the remark in Part I, Section 5.3, that the root mean square "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp": over the reliable range the root mean square of $(T-M)/\sqrt{N}$ decreases like $(mm')^{-3/8}$, and the second term $N^{1+\varepsilon}/(\det A)^{1/4}$ of Waibel's Theorem 1 appears to overestimate $\|G_{m,m'}\|^2$ by a factor of about $(mm')^{1/4}$ for this family. Figure 1(a) shows that the estimate has reached its plateau for the five pairs (the running estimate is constant for $y \geq 10L$), and that $(1009, 1013)$, with $Y/L \approx 0.5$, is still in the transition region, where Theorem 1.1 only gives the bound $Y\|G\|^2$; Figure 1(b) shows the five reliable points on the line of slope $1/4$.

We record what a norm bound of this strength would give, as a conditional statement.

Hypothesis 5.2. $\|G_{m,m'}\| \ll_\varepsilon (mm')^{1/8+\varepsilon}$ for all coprime odd m, m'.

Proposition 5.3. *Under Hypothesis 5.2: Theorem 2.3 and Corollary 2.7 of Part I hold with* $(mm')^{1/8+\varepsilon}$ *in place of* $(mm')^{1/4+\varepsilon}$*; the level of distribution of* $r(n)r(N-n)$ *is* $4/9$ *for every* $N \equiv 2 \pmod 4$ *and* $4/5$ *for almost all such* N*; and the sieve of Part I gives* $\Omega(n(N-n)) \leq 8$ *and* $\max(\Omega(n), \Omega(N-n)) \leq 6$ *for every* N*.*

Proof. The proofs of Theorem 2.3 and Corollary 2.7 of Part I use the norm only through Lemma 2.1. In Theorem 3.2 of Part I the exponent $1/4$ enters through $\sum_{k \leq D} d(k) k^{1/4} \ll D^{5/4} N^\eta$; with $1/8$ the balance $N^{1/2} D^{9/8} = N$ gives $D = N^{4/9}$ (Remark 3.3 of Part I), and in Theorem 1.2 the balance $X^{3/2} D^{5/8} = X^2$ gives $D = X^{4/5}$. The floating-point scan at $\theta = 4/9$ gives, at best, $1/\lambda + 2/\beta_1 \approx 8.61$ and $1/\lambda + 1/\beta_1 \approx 5.98$. The second value, however, is attained only at the largest admissible λ, where $L - \lambda R = 0$; any λ with a positive margin $L - \lambda R > 0$ pushes $1/\lambda + 1/\beta_1$ above 6. Hence $(8, 6)$ is attainable with margin and can be certified as in Part I, while $(8, 5)$ is not reachable with these weights at this level; we claim only $(8, 6)$. For almost all N the level $4/5$ does not improve on Corollary 1.3: the scan at $\theta = 4/5$ gives at best $1/\lambda + 2/\beta_1 \approx 6.15$ and $1/\lambda + 1/\beta_1 \approx 4.22$, i.e. again $(6, 4)$; the Richert weights with one parameter β_1 no longer profit from the level beyond $2/3$, and a lower bound such as $(5, \cdot)$ would need different weights. Figure 2 shows the two quantities as functions of θ. □

Hypothesis 5.2 concerns the norm alone and is a statement about a single theta series, not about its individual coefficients; it might be more accessible than Conjecture 5.1 of Part I. Waibel's lower bound (Theorem 3 of [3]) gives only $\|G_{m,m'}\|^2 \gg 1/\min(m, m') + O(L^\varepsilon)$ for these forms and is far from the truth here; the growth $(mm')^{1/4}$ of $\|G\|^2$, if real, asks for an explanation in terms of the spectral weights $\langle \Theta, \varphi \rangle$. By Parseval over the orthogonal eigenbasis $\mathcal{B}^*$,

$$\|G\|^2 = \sum_{\varphi \in \mathcal{B}^*} \frac{|\langle \Theta, \varphi \rangle|^2}{\|\varphi\|^2},$$

so Hypothesis 5.2 is the statement that this weighted second moment over the family of $\asymp L$ eigenforms is $\ll L^{1/4+\varepsilon}$. In the heuristic factorisation of Part I, Section 5.5, $\langle \Theta, \varphi \rangle = \mathcal{E}_\varphi(m, m') L(\tfrac{1}{2}, f) L(\tfrac{1}{2}, f \otimes \chi)$, and $\|G\|^2$ becomes a second moment of products of two central values with the weights $|\mathcal{E}_\varphi|^2/\|\varphi\|^2$. (Since θ^2 is an Eisenstein series, the inner product is a Rankin–Selberg integral, not a triple product of cusp forms.) Without the local factors $\mathcal{E}_\varphi$, which have not been worked out, this does not predict the exponent $1/4$; it does locate the question in the second moment of a family of L-values, where tools exist. The factors $\mathcal{E}_\varphi(m, m')$ come from two sources: the expansion of $\theta^2(m'z)$, a weight-one Eisenstein series of level $4m'$ and character χ_{-4}, in the Eisenstein series attached to the cusps of $\Gamma_0(4m')$ (this is where the unfolding takes place), and the oldform structure of φ, i.e. its coordinates in the basis $B_t f$ of Part I, Lemma 2.2. Writing them out explicitly is the first concrete step towards Hypothesis 5.2.

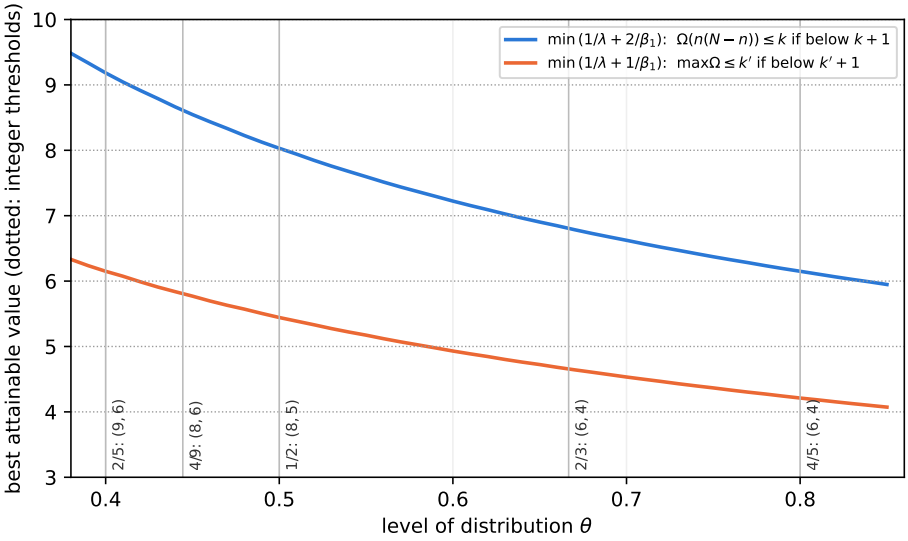


FIGURE 2. The smallest values of $1/\lambda + 2/\beta_1$ and of $1/\lambda + 1/\beta_1$ attainable with $L - \lambda R > 0$ in the sieve of Part I, as functions of the level of distribution θ (floating-point model of `modular_sieve_scan.py`, minimised over α, β_1 with $\lambda = L/R$; script `plot_sieve_thresholds.py`). The bound $\Omega(n(N-n)) \le k$, resp. $\max \Omega \le k'$, holds when the blue, resp. orange, curve is below the dotted line $k+1$, resp. $k'+1$. The vertical lines mark $\theta = 2/5$ (Part I), $4/9$ and $4/5$ (Proposition 5.3), $1/2$ (Conjecture 5.1 of Part I) and $2/3$ (Corollary 1.3); only the parameter sets at $2/5$, $1/2$ and $2/3$ are certified. Between $2/3$ and $4/5$ both curves stay just above the thresholds 6 and 4.

DATA AND CODE

The scripts `mean_square_check.py`, `certify_modular.py` (argument `2/3`), `plot_mean_square.py` and `plot_sieve_thresholds.py` (Figures 1 and 2), the modules of Part I they use (`sup_error_scan.py`, `modular_sieve_scan.py`, `sieve_core.py`) and the outputs `results/mean_square_check_x21.txt`, `results/certify_modular_output_twothirds.txt`, `results/sieve_thresholds.txt` are in the repository https://github.com/Ruqing1963/two-squares-mean-square; the code and data of Part I are at https://github.com/Ruqing1963/two-squares-theta-level. The certification of the sieve parameters uses directed rounding as described in Part I, Appendix A; the mean-square computation is in floating point and is evidence, not proof.

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